What makes motion hard to stop?

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Compare momentum changes in a controlled stopping situation.

SUCCESS CRITERIA - I can...

1. Compare mass and velocity while controlling other variables.
2. Use momentum change to compare stopping force for equal times.
3. Avoid claiming that momentum alone determines damage.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “momentum” and “impulse” to compare your model or result. Sentence starters are optional.

momentum: mass times velocity, including direction

impulse: net force acting over time; change in momentum

system: the object or objects chosen for analysis

01 Bring back an earlier idea

Two ideal Sun-orbits have $a_B = 3\ a_A$.
Find T_B/T_A.
Explain why the same normalized Sun-orbit equation is not automatically valid around another body.

Word help: orbit = a path around another body under gravity

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $p = mv$; $F_{avg} = \Delta p/\Delta t$.
Momentum p includes direction. For a fixed stopping time, greater momentum change needs greater average net force.

Worked reasoning: A 2 kg cart and a 4 kg cart move right at 3 m/s. Their momenta mv are 6 and 12 kg·m/s. Stopping the second requires twice the momentum change.

Your turn: Draw the velocity and momentum arrows for both carts.
Explain which pair has equal lengths.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: ① ② ③

03 Represent and interpret

Carts A and B both move right at 2 m/s. A has mass 1 kg; B has mass 3 kg. Which has more momentum?
State the controlled quantity.

Sentence starter: I notice ... . This supports ... because ... .



A: 1 kg; 2 m/s B: 3 kg; 2 m/s

Marzano check - circle: ① ② ③

04 Solve and justify

Compare A:(1 kg,+2 m/s), B:(3 kg,+2 m/s), C:(1 kg,+4 m/s). Each stops in .50 s.
Find signed changes in momentum and average forces.

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Analyze the reasoning

Two objects have equal momentum but strike different materials over different times. Can equal momentum alone establish equal damage?
Explain.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3

06 Choose a method

Two equal 4000 kg masses are 6 m apart.
Use $G=6.67\times10^{-11}$ N·m²/kg².
Find force. By what factor is an answer wrong if r rather than r² is used?

Sentence starter: The step I would change is … . The evidence is … .

Marzano check - circle: 1 2 3

07 Exit check

A 3 kg cart moving right at 4 m/s stops in 4 s.
Find signed impulse and average force with right positive.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

A 0.5 kg cart at +8 m/s and a 2 kg cart at +2 m/s each stop in 0.2 s.
Compare their momentum changes and average stopping forces.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3